

The Development of an Efficient System to Monitor the Honeybee Colonies Depopulations

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Abstract

The world of honeybees (*A. mellifera*) are facing dramatic changes. Since the late 1990s, beekeepers around the world have observed mysterious and sudden disappearance of bees, and reported unusually high rates of decline in honeybee colonies. In the same time, the numerous studies show also a decline of other pollinators' populations. The main causes seldom incriminated are climate change, pesticide use, habitat loss, parasites, but looking at real issues leading to honeybees' populations reduction, but some french studies show that major problems are pesticides (52%), Asian hornet (*Vespa velutina*), other diseases and pests (i.e. *Varroa destructor*) (32%), weather (16%), thief (10%). Taking into account that honeybees are a very good environmental biosensor, the development of an advanced electronic system that assess the development of population status at any moment of the season is very important for environmental quality evaluation and for beekeeping management, to evaluate the risks and prevent depopulation phenomenon. Therefore, the present study aims to develop a unique connected advanced solution for beehives to give direct information to beekeepers on pesticide contamination suspicion, hornet attack detector, swarming signals and thief alarm. Two major features, as compared to market competitors, were developed on the new system: a device to count forager bees, therefore get many information about the entrance traffic of honeybees and an advanced sound analysis to detect hornet, but also other signals connected with specific physiological status of honeybee colonies.

Key words: honeybee, monitoring systems, sensors, forager workers, hornet.

I. INTRODUCTION

The advanced technologies and the actual stage of informatics conquered almost all the domains including honeybees [1, 8, 9]. Internet of Things (IoT) represents the newest concept of collecting information from the environment, natural or artificial, in order to assess some risks or to have a better control and management. The honeybee colony is, from this point of view, a natural complex system, integrated in the natural environment, which gives important information about a large area, taking into account its specific

behavioural characteristics of forage flight and food collection. Thus, the honey bee can be considered a true biosensor of the natural environment. The importance of honeybees for pollination services in natural ecosystems and its decline in the last decades was very well highlighted in numerous studies [2, 6, 7]. The collected data from a honeybee colony could give information about its physiological status which can be correlated with the evolution of some environmental factors: temperature, humidity, floral resources, thus obtaining information about the climatic and vegetation conditions as well as the quantity and quality of food.

Following these, about 20 companies develop and commercialize connected hives solutions in different countries: Simbee (Romania), Buzzbox (USA), Hostabee (France), Beezon (Greece), Arnia (UK), IoBee (EU project), etc.

Generally, all these sensor hives are equipped with:

- Weight sensor
- Internal and external temperature sensors
- Internal and external humidity sensor

The weight parameter gives information about the honey flows evolution, the degree of flowering and even pollination management. This parameter is very important for beekeeper as it helps to optimize the flora valorisation and transhumance [4, 5].

The external and internal temperature and humidity parameter could be correlated with the conditions of vegetation and melliferous flora as well as nectar secretion.

So, generally, all these data offer information about:

- The evolution of honey flows, so that, beekeepers to take actions about the apiary movement, necessity of artificial feedings or honey extraction.
- The evolution of environmental factors in order to understand their impact on nectar secretion which is directly correlated with honey production.
- The evolution of the honeybee colonies development, both in active and inactive seasons [4, 5].

A specific data base registers and processes all the measured data in order to be graphically expressed and establish the evolution of different parameters. The parameters are registered at short intervals of time in the day, and can be periodically, sent to the user, in order to alert and inform in real time (SMS).



Fig. 1. A sensor beehive equipped with weight, temperature and humidity control sensors.

For example, the monitoring hive, gives to the beekeeper, rapidly, the possibility to be informed about the honey flow and take action in case of unexpected values

Beside these practical aspects, these hives represent an important research tool, especially by analyzing the evolution of the parameters in specific fields or comparing data evolution in different years.

In some experiments [3], the system was used to monitor the honey flows of some cultures (rape, sunflower) in order to have indications about pesticides (neonicotinoids) contaminations.

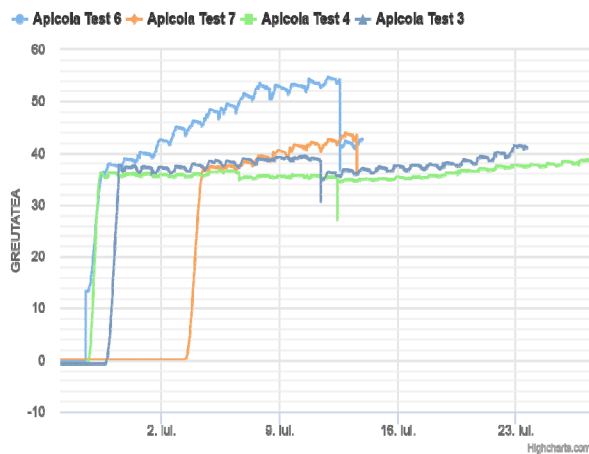


Fig. 2. The evolution of honey flows, by weight gain, in four different locations of sunflower fields, in Romania, measured with Simbee system.

The colony activity in the fields was correlated with laboratory analyses for honey type's identification in order to establish the monofloral targeted samples, which to be relevant for the neonicotinoids analyses of a specific culture. The food (nectar and pollen) income, by means of hive weight registration, gives the first indications about the honey/pollen flow or about the forager bees' activity.

To better establish any depopulation of the monitoring hive in the field it is necessary to develop a system of evaluation and processing the entrance traffic of the foragers' bees (out and in the hive), in order to establish with a higher accuracy, the depopulations of honeybee colonies, especially in the context of pesticide use and hornet attack.

The traffic sensor/device could help beekeeper to establish the toxicity of food sources from bee flight area which conduct to depopulation/mortalities.

The noise sensor/device could offer information about the bee colony physiological status (swarming signals, colony death, and presence of queens) and hornet presence in the proximity area of hive.

II. LIMITATION OF THE CURRENT TECHNOLOGY APPROACH

As mentioned in introduction chapter, two main categories of connected hives have developed in the last years. The ones that allow the beekeeper to follow the weight of the beehive, and the ones that allow to follow the temperature and humidity of the beehive. To put it in a nutshell, it allows to monitor the hive level of nectar harvested, honey readiness, a certain, imprecise, status of the colony development. However, it does not allow to cope with bees' threats like the degree of depopulations, especially in sublethal effects of pesticides cases, hornet's attack, contamination, varroosis depopulations. Beyond, current prices on the market prevent adoption by beekeepers. A weighting scale is worth several hundreds of euros. Such a price for a single device is not compatible with beekeepers' revenues. To develop cheaper solution, together with a higher accuracy of depopulation measurement in a connected hive system is an challenging, but an important goal. Contribution of the technology is powerful when it is at scale, i.e.

- when a large and varied amount of a population is involved. The technology is then participative. It may be simple and concrete like allowing to monitor the progress of the Asian hornet across Europe for instance. Each time a beekeeper or any citizen sees an Asian hornet, it pushed a button on an application and the position is automatically geo-located thanks to the smartphone GPS.
- when the data collected is wide enough to make a big data and feed Artificial Intelligence. Beyond the service of monitoring or alerting for the beekeeper, the system gathers and treat data for research.

National or private Institutes of Research have deeply studied the bees and developed knowledge. Technology, through an easy to use and affordable device, would add the power of the mass of data to the deep knowledge of the researchers. A cooperation between AgTech start-ups and the Research world

would lead to the next level on bees understanding. As long as technology is designed to collect data

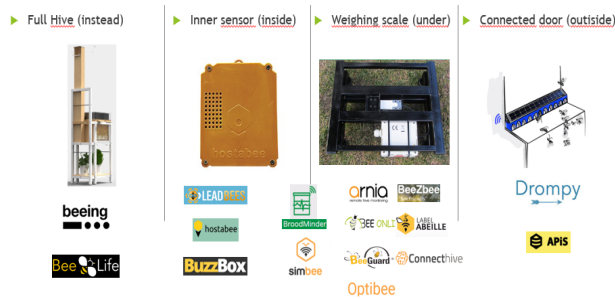


Fig 3. Examples of start-ups in the connected beehives market.

Another approach is the native connected beehive that consists in deeply integrated solutions to take care of the bees. It claims to be even designed to harvest honey without being in contact with bees. It focuses on a public of novice, mainly urbans, seeking to get a hive without having to manipulate bees.

Finally, is the connected door. Thanks to the features of bees counting, it focuses on threats of the bees. It alerts in case of suspicion of contamination of a hive, since it detects a gap between the bees going out and the bees coming back. The device can also warn the beekeeper as soon as a swarm leaves the hive and allow him to come and collect it before it goes away in the nature. Last the connected door alerts in case of hornets and Asiatic hornets' attacks.

Asiatic hornet has been introduced in Europe first in the south west of France at the occasion of an import of pottery from China in 2004. Therefore, France beekeepers has been exposed early to the damages of this new predator. It has now reached neighbouring countries and there is no reason that it does not manage to settle in the whole Europe, except mountains regions basically, in the coming years. Two figures illustrate the speed of expansion of this species:

- up to 75 thousand nests has been destroyed in 2018 in France
- In Paris area, 500 Asiatic hornets' colonies were killed in 2017 versus 2500 in 2018.

France had to develop research about Asiatic hornet to understand this new threat. Asiatic hornet threat goes beyond beekeeping impact. A species is considered as invasive when it creates one of the following issues: an ecological issue, an economic issue, or human health issue. Asiatic hornet matched the 3 pillars. Ecologically, it consists in a new predator hardly regulated by existing local species apart from few birds like the great spotted woodpecker, chickadees, crows or European bee-eater, or the parasite fly *Conops vesicularis*. From an economic point of view, it impacts the beekeeping activity but not only, the wine-growing appears seriously damaged too as it attacks the grapes as a source of sugar. Last but not least

Asiatic hornets impact human health since their sting can even be lethal.

III. THE ISSUE FACED BY BEEKEEPING WILL NOT BE SOLVED IF AGRICULTURE PRACTICES DOES NOT EVOLVE

Agriculture practices directly affects bee's health in the surrounding. The two topics are closely linked. Beyond the bees, the whole biodiversity is at stake here. The society concentrates its debates on organic or vegan when it comes to agriculture. Nevertheless, soil biodiversity may be one of the major stakes of our time. Lately, studies and agricultural on field trials showed the importance of a living soil in agricultural productivity. Worms are representatives of the symbiosis between plants and the soil biodiversity. A good fertile soil may show up to 60% of aeration thanks mainly to anecic earthworms. Plants use the galleries to deeply and quickly deploy its roots in the soil and access water reserve and nutrition. Worms are also responsible for the creation of the clay-humus complex, which is electrically attached, and so resist to bad weathers compared to deteriorated soils.

The heavy use of chemistry and machines in the agriculture was understandable in the last century to answer the need of food supply. Nowadays it shows its side effects, especially on bees and biodiversity, but also on long term productivity, human health or even natural disasters like mudslides. The E.U. "greened" the Common Agricultural Policy with subsidies linked to green practices criteria. The green grants to farmers represent billions of spending. The COP21 in France set up the *Ifor1000* objectives to transform agriculture as the most powerful tools to success in the ecological transition. If agriculture manages to stock 0,1% of carbon each year in the soils thanks to agro-ecological practices, it would balance for over-emission by the other human activities.

Technology has to play its role. First to assist the farmers in the transition towards sustainable practices, they will need data about their soils, and will need to understand the impact of their practices on the soil. Second, on an ecological point of view, to understand the evolution of the life and biodiversity of the soils, in real-time and all over the territories. Sensors are under development with such objectives. It would complete the bees data from connected beehives. Using bees and biologic life of the soils as bio-indicators could allow to draw maps of the territories according to their biodiversity quality. How long U.E. CAP green shift can be maintained if we are not able to measure the impact of the billions invested?

IV. CARRYING OUT NEW R&D

Following the previous introduction, there was a need for new R&D for a totally new and low-cost product. The aim of this new research is to develop a unique connected advanced solution for beehives to give direct information to beekeepers on pesticide contamination suspicion, the level of development of the bee colony population throughout the year, hornet attack detector, swarming signals and thief alarm.

In order to achieve the main objectives of the project the following stages were established:

1. To design and adapt to hive the new components of bee

colony monitoring into a CONNECTED ADVANCED SOLUTION to a device:

- bees counter (to count forager bees),
 - an advanced sound analysis (for hornet attack and swarming signals), and
 - an alarm system (against theft);
2. To develop the data COMMUNICATION system. Advance communication part with low energy long distance (e.g. LORA) which to include radio optimisation (e.g. antenna);
 3. To test a PROCESSOR (a low energy and processing module);
 4. To optimize a HARVESTER solution (Energy optimisation and harvester, focused on an autonomous system - a small solar panel) ;
 5. To develop a SOUND ANALYSIS system (a very advance sound analysis module). The microphones (such as MEMS) to detect bees sounds as well as specific hornet sound. (PoC has proven such feasibility). The studies will focus on microphones positions and processing capabilities with the limited energy;
 6. ADD-ONS Additional part adding MEMS accelerometer for thief detection and other parameters such as temperature and humidity;
 7. The experimental tests in laboratory conditions;
 8. The experimental tests in field conditions, on honeybee colonies.

A series of MVP (Minimum Viable Products) will be developed within 2019-2020 thanks to the DIATOMIC project. We will validate the product with beekeepers thanks to R&D partner, ICDA working with 20K beekeepers. We plan to hit the market early 2020 with a target price between 100 and 150 euros for a full autonomous system.

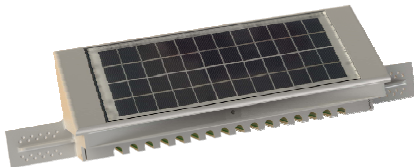


Fig 4. 3D view of the expected autonomous connected door

V. EXPECTED BENEFITS AND IMPACTS

By achieving the new monitoring system we expect having the following benefits:

- **Monitor health & location:** Thanks to embedded real-time devices in the beehives, the beekeepers are notified early of any anomaly, allowing them to react early to critical health threats (abnormal activities, hornet, humidity, temperature) or theft (accelerometer/GPS movements triggers alert). Expected benefit: 50% increase product after 1 year;

- **Strengthen colonies:** Based on the collected data and regular on-site visits, our veterinary advisors can train and administer best practices early and effectively;
- **Grow & increase productivity:** Thanks to above factors, our average beehive duplicates with a rate of 75% (vs market avg of 40%) and honey productivity could improve a lot;
- **Meet loan criteria:** Many institutional investors such as banks, microfinance institutions have high demands concerning and risk management tools. Our solution gives beekeepers the chance to meet these demands and facilitate funding for growth;
- **Produce higher quality:** With early and concise treatment the bees are only minimally exposed to medication and sugar feeding, thus producing a very pure honey. Expected benefit: Sampling of honey will demonstrate increase of quality.

As for the environmental impact, it goes without saying that the beekeeping sector is an integral part of European agriculture. While something over 620,000 EU citizens receive their main income or additional earnings from this sector, its importance is much greater than judged from the amount of its contribution to the gross production value of the economy. According to EU reports, 84% of plant species and 76% of Europe's food production depend on pollination by bees, so that the economic value thus produced – estimated at EUR 14.2 billion annually in the EU – exceeds the value of the honey produced by a long way [6]. Having in mind the relevance of bees for the overall population, by collecting relevant data timely and accompanied with veterinary extension service, our new product will directly contribute to (UN Sustainable Development Goals) SDG 2 in a sense of achieving food security and promotion of sustainable agriculture. More over, the project contributes to ensuring sustainable food production systems and implementing resilient agricultural practices that increase productivity and production, that help maintain ecosystems, and that strengthen capacity for adaptation to climate change.

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